



DUQ: Dual Uncertainty Quantification for Text-Video Retrieval

Xin Liu¹, Shibai Yin^{1*}, Jun Wang^{1*}, Jiaxin Zhu¹, Xingyang Wang¹ and Yee-Hong Yang²

¹Southwestern University of Finance and Economics

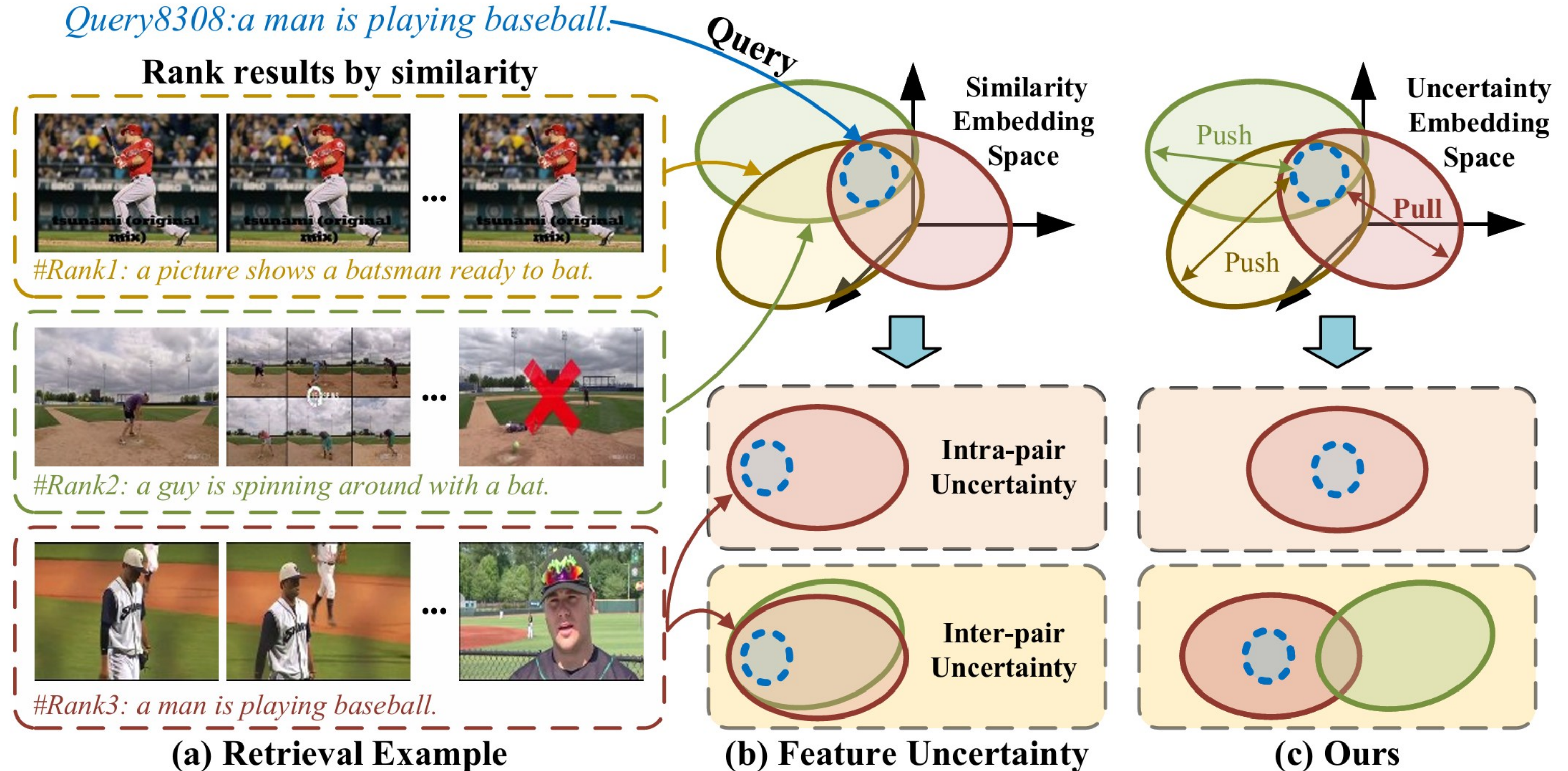
²University of Alberta



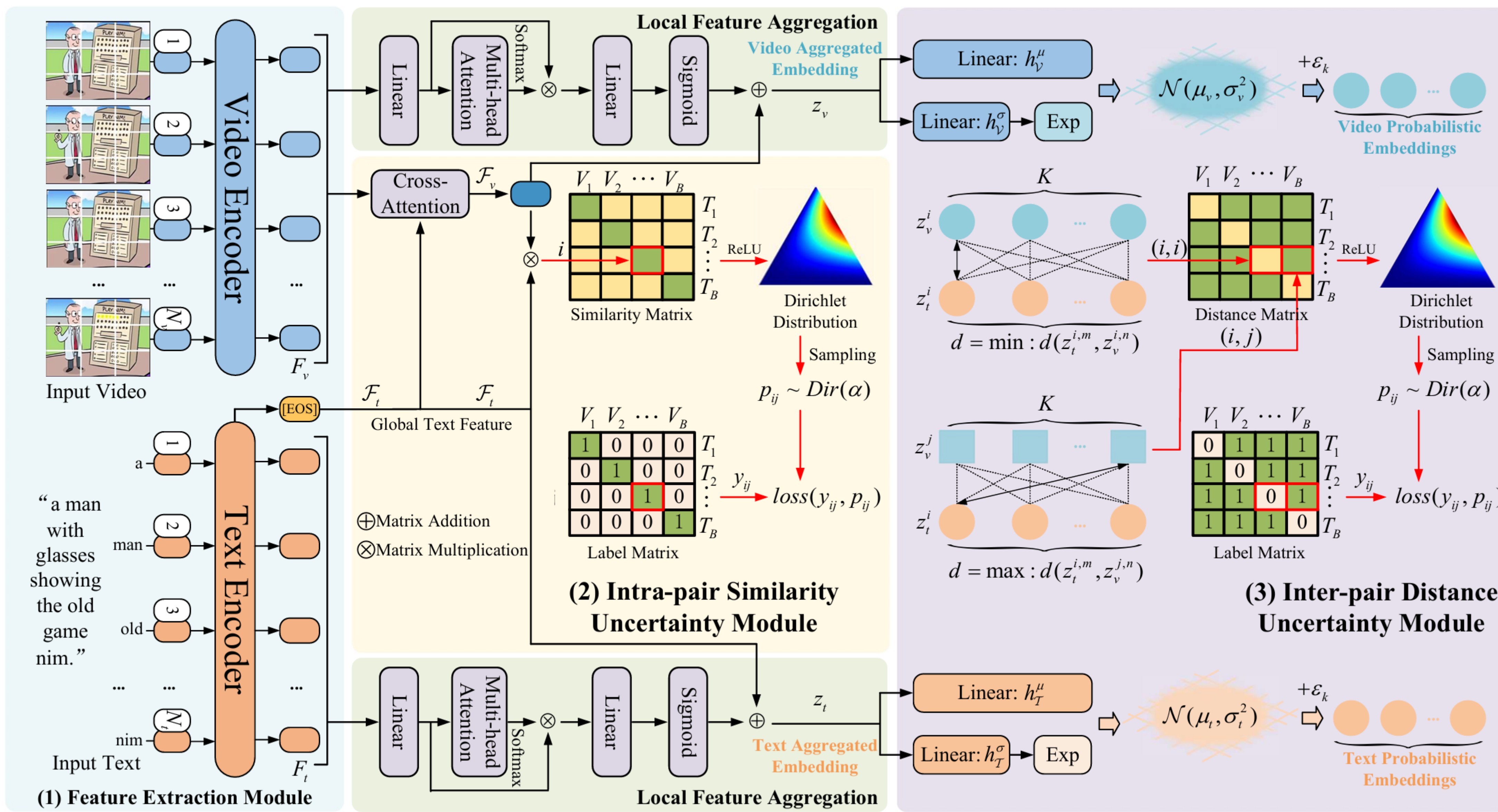
Abstract

- Text-video retrieval establishes accurate similarity relationships between text and video through granularity alignment and feature enhancement. However, relying solely on similarity to associate intra-pair features and distinguish inter-pair features is insufficient, e.g., when querying a multi-scene video with sparse text or selecting the most relevant video from many similar candidates.
- In this paper, we propose a novel Dual Uncertainty Quantification (DUQ) model that separately handles uncertainties in intra-pair interaction and inter-pair exclusion. Specifically, to enhance intra-pair interaction, we propose an intra-pair similarity uncertainty module to provide similarity-based trustworthy predictions and explicitly model this uncertainty. To increase inter-pair exclusion, we propose an inter-pair distance uncertainty module to construct a distance-based diversity probability embedding, thereby widening the gap between similar features.
- We evaluate our model on six benchmark datasets: MSRVT (51.2%), DiDeMo, LSMDC, MSVD, Charades, and VATEX, achieving state-of-the-art retrieval performance.

Motivation



Method



1. Baseline (FEM)

$$s(t, v) = \frac{\mathcal{F}_t \cdot \mathcal{F}_v}{\|\mathcal{F}_t\| \cdot \|\mathcal{F}_v\|}$$

$$\mathcal{L}_S = -\frac{1}{2} \left(\frac{1}{B} \sum_{i=1}^B \log \frac{e^{s(t_i, v_i) \cdot \lambda}}{\sum_{j=1}^B e^{s(t_i, v_j) \cdot \lambda}} + \frac{1}{B} \sum_{i=1}^B \log \frac{e^{s(t_i, v_i) \cdot \lambda}}{\sum_{j=1}^B e^{s(t_j, v_i) \cdot \lambda}} \right)$$

2. Intra-pair Interaction (ISUM)

$$\mathbb{E}_{p_{ij} \sim \text{Dir}(\alpha_{ij})} \|y_{ij} - p_{ij}\|_2^2$$

$$\mathcal{L}_i^U = \sum_{j=1}^B (y_{ij} - \hat{p}_{ij})^2 + \frac{\hat{p}_{ij}(1 - \hat{p}_{ij})}{S_i + 1}, \hat{p}_{ij} = \mathbb{E}[p_{ij}] = \alpha_{ij} / S_i$$

$$\mathcal{L}_S^U = \frac{1}{B} \left(\sum_{i=1}^B \mathcal{L}_i^U + \sum_{j=1}^B \mathcal{L}_j^U \right)$$

3. Inter-pair Exclusion (IDUM)

$$\begin{cases} p(z|t) \sim \mathcal{N}(h_t^\mu(z_t), e^{h_t^\sigma(z_t)}) \\ p(z|v) \sim \mathcal{N}(h_v^\mu(z_v), e^{h_v^\sigma(z_v)}) \end{cases} \begin{cases} z_t^k = h_t^\mu(z_t) + e^{h_t^\sigma(z_t)} \cdot \epsilon_k \\ z_v^k = h_v^\mu(z_v) + e^{h_v^\sigma(z_v)} \cdot \epsilon_k \end{cases}$$

$$d(z_t^i, z_v^j) = \begin{cases} \max d(z_t^{i,m}, z_v^{j,n}), & \forall n, m, i \neq j, \\ \min d(z_t^{i,m}, z_v^{j,n}), & \forall n, m, i = j. \end{cases}$$

$$\mathcal{L}_D = \frac{1}{2} \left(\frac{1}{B} \sum_{i=1}^B \log \frac{e^{d(z_t^i, z_v^i) \cdot \lambda}}{\sum_{j=1}^B e^{d(z_t^i, z_v^j) \cdot \lambda}} + \frac{1}{B} \sum_{i=1}^B \log \frac{e^{d(z_t^i, z_v^i) \cdot \lambda}}{\sum_{j=1}^B e^{d(z_t^j, z_v^i) \cdot \lambda}} \right)$$

$$\mathcal{L}_D^U = \frac{1}{B} \left(\sum_{i=1}^B \mathcal{L}_i^U + \sum_{j=1}^B \mathcal{L}_j^U \right)$$

Experiments

Table1: Retrieval performance on the MSRVT-1K datasets.

Methods	MSRVT (Text-to-Video)					MSRVT (Video-to-Text)				
	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
CLIP-Non	-	-	-	-	-	-	-	-	-	-
CE [Liu et al., 2019]	20.9	48.8	62.4	6.0	28.2	20.6	50.3	64.0	5.3	25.1
MMT [Gabeur et al., 2020]	26.6	57.1	69.6	4.0	24.0	27.0	57.5	69.7	3.7	21.3
SST [Patrick et al., 2020]	27.4	56.3	67.7	3.0	-	26.6	55.1	67.5	3.0	-
CLIP-ViT-B/32	-	-	-	-	-	-	-	-	-	-
EMCL-Net [Jin et al., 2022]	46.8	73.1	83.1	2.0	-	46.5	73.5	83.5	2.0	-
X-Pool [Gorti et al., 2022]	46.9	72.8	82.2	2.0	14.3	44.4	73.3	84.0	2.0	9.0
DiCoSA [Jin et al., 2023b]	47.5	74.7	83.8	2.0	13.2	46.7	75.2	84.3	2.0	8.9
UATVR [Fang et al., 2023]	47.5	73.9	83.5	2.0	12.3	46.9	73.8	83.8	2.0	8.6
HBI [Jin et al., 2023a]	48.6	74.6	83.4	2.0	12.0	46.8	74.3	84.3	2.0	8.9
Cap4Video [Wu et al., 2023]	49.3	74.3	83.8	2.0	12.0	47.1	73.7	84.3	2.0	8.7
T-Mass [Wang et al., 2024]	50.2	75.3	85.1	1.0	11.9	47.7	78.0	86.3	2.0	8.0
DUQ (Ours)	51.2	77.3	86.1	1.0	10.8	50.4	79.2	87.5	1.0	6.4
CLIP-ViT-B/16	-	-	-	-	-	-	-	-	-	-
X-Pool [Gorti et al., 2022]	48.2	73.7	82.6	2.0	12.7	46.4	73.9	84.1	2.0	8.4
UATVR [Fang et al., 2023]	50.8	76.3	85.5	1.0	12.4	48.1	76.3	85.4	2.0	8.0
Cap4Video [Wu et al., 2023]	51.4	75.7	83.9	1.0	12.4	49.0	75.2	85.0	2.0	8.0
T-Mass [Wang et al., 2024]	52.7	77.1	85.6	1.0	10.5	50.9	80.2	88.0	1.0	7.4
DUQ (Ours)	55.9	81.0	88.6	1.0	8.4	54.6	82.4	89.9	1.0	5.3

Table2: Retrieval performance on other datasets.

Methods	DiDeMo (Text-to-Video)					LSMDC (Text-to-Video)				
	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
CE [Liu et al., 2019]	16.1	41.1	-	8.3	43.7	11.2	26.9	34.8	25.3	-
EMCL-Net [Jin et al., 2022]	45.3	74.2	82.3	2.0	12.3	23.9	46.4	53.7	8.0	-
TS2-Net [Liu et al., 2022]	41.8	71.6	82.0	2.0	14.8	23.4	42.3	50.9	9.0	56.9
X-Pool [Gorti et al., 2022]	44.6	73.2	82.0	2.0	15.4	25.2	43.7	53.5	8.0	53.2
CLIP-VIP [Xue et al., 2022]	48.6	77.1	84.4	2.0	-	25.6	45.3	54.4	8.0	-
DiCoSA [Jin et al., 2023b]	45.7	74.6	83.5	2.0	11.7	25.4	43.6	54.0	8.0	41.9
DiffusionRet [Jin et al., 2023c]	46.7	74.7	82.7	2.0	14.3	24.4	43.1	54.3	8.0	40.7
UATVR [Fang et al., 2023]	43.1	71.8	82.3	2.0	15.1	-	-	-	-	-
DUQ (Ours)	51.8	77.9	86.5	1.0	10.6	28.5	48.2	58.0	6.0	41.2
Methods	Charades (Text-to-Video)					VATEX (Text-to-Video)				
	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
Support-Set [Patrick et al., 2020]	-	-	-	-	-	44.9	82.1	89.7	1.0	-
CLIP4Clip [Luo et al., 2022]	9.9	27.1	36.8	21.0	85.4	-	-	-	-	-
X-Pool [Gorti et al., 2022]	11.2	28.3	38.8	20.0	82.7	60.0	90.0	95.0	1.0	3.8
UATVR [Fang et al., 2023]	-	-	-	-	-	61.3	91.0	95.6	1.0	3.3
T-Mass [Wang et al., 2024]	14.2	36.2	48.3	12.0	54.8	63.0	92.3	96.4	1.0	3.2
DUQ (Ours)	28.5	55.0	66.8	4.0	22.9	80.0	97.4	99.0	1.0	1.6

Table3: Ablation study of different components.

Baseline	DUQ	MSRVT (Text-to-Video)					MSRVT (Video-to-Text)				
		R@1↑	R@5↑	R@10↑	MdR↓	MnR↓	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
1/✓		46.9	74.5	82.2	2.0	14.3	44.4	73.3	84.0	2.0	9.0
2/✓	✓	47.5	73.9	83.5	2.0	12.3	46.9	73.8	83.8	2.0	8.6
3/✓	✓	48.2	74.6	83.4	2.0	12.2	45.2	73.7	84.8	2.0	8.6
4/✓	✓	49.2	77.6	85.4	2.0	11.2	49.2	77.7	84.8	2.0	8.1
5/✓	✓	48.9	76.3	84.6	2.0	12.6	48.5	74.5	84.3	2.0	8.1
6/✓	✓	51.2	77.3	86.1	1.0	10.8	50.4	79.2	87.5	1.0	6.4

Visualization



Conclusion

In this paper, we propose a novel Dual Uncertainty Quantification (DUQ) framework to provide trustworthy predictions by quantifying the confidence of intra-pair interactions and inter-pair exclusions induced by data uncertainty. Accurate text-video retrieval requires reliable intra-pair associations and inter-pair exclusions to ensure the model effectively captures semantic consistency between text and video. Extensive experiments on six benchmark datasets demonstrate that our approach achieves new SOTA retrieval performance. We hope this work inspires further applications in other domains.

Link

- Homepage: <https://github.com/OPA067>
- DUQ Code: <https://github.com/OPA067/DUQ>
- You may also be interested in:
- T-Text: Temporal Text Representation Learning for Text-Video Retrieval. <https://github.com/OPA067/T-Text>
- HVD: Human Vision-Driven Video Representation Learning for Text-Video Retrieval. <https://github.com/OPA067/HVD>
- Spot the Key, Recover the Rest: Dual-Stage Cross-Modal Learning for Text-Video Retrieval. <https://github.com/OPA067/SKRR>

